TING-LUN SU

Intelligent Automation Engineer

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Location: Taichung, Taiwan

EDUCATION

B.E. in Intelligent Automation Engineering
National Chin-Yi University of Technology (NCUT), Taichung, Taiwan
June 2022 - Present
Focus: AI-driven robotics, intelligent manufacturing, computer vision, and automation systems

RESEARCH INTERESTS

- Intelligent Manufacturing & Industry 4.0
- AI-Driven Robotic Systems & Path Optimization
- Computer Vision & Quality Inspection
- Large Language Models (LLM) & Multi-modal AI Applications
- Human-Robot Collaboration & Smart Automation

PUBLICATIONS

Journal Papers

T.-L. Su, P.-J. Chen, T.-J. Peng, and C.-C. Lin, "Automated spray path planning based on Bayesian optimization and ant colony optimization," The International Journal of Advanced Manufacturing Technology, vol. 139, pp. 5491-5509, Aug. 2025.
DOI: 10.1007/s00170-025-16162-x | SCI-indexed, Q2, Impact Factor: 2.9

T.-J. Peng, P.-J. Chen, T.-L. Su, and C.-C. Lin, "Optimization of Automated Forging Release Agent Spray Path Based on Ant Colony Algorithm," Journal of Automation Intelligence and Robotics, TAIROA, No. 51, pp. 114-120, Dec. 2024.

Conference Papers

Z.-L. Sun, T.-S. Liaw, C.-L. Liu, T.-L. Su, C.-C. Wang, and T.-J. Peng, "Green Machining Optimization and Burr Treatment in Mill-Turn Operations," in 27th Annual Meeting of the Chinese Society of Mechanism and Machine Theory (CSMMT), Nov. 2024, Abstract No. 039.

T.-L. Su, C.-H. Du, and C.-H. Lai, "Development of a Contactless AIoT Smart Data Management Hybrid Vending System," in The 17th Intelligent Living Technology Conference (ILT), Jun. 2023, Paper No. 3C009.

T.-L. Su, Q.-L. Liu, Y.-H. Lin, and C.-H. Lai, "The function analysis of green electric automatic weeder," in Conference on Green Technology Engineering and Applications (GTEA), May 2023, Paper No.

GEES012, pp. 214.

PATENTS

Invention Patent: Mushroom Stem Cutting Machine
Application No.: I891383 (Taiwan), May 2025
An automated device designed to precisely cut mushroom stems, improving efficiency and consistency in agricultural processing.

Utility Model Patent: Mushroom Stem Cutting Machine
Application No.: M659734 (Taiwan), August 2024
A mechanical design enhancing the stability and cutting performance of the mushroom stem processing system.

INDUSTRY-ACADEMIA COLLABORATION PROJECTS

Visual Measurement Inspection Station and Generative AI Navigation Assistance System
TURVO INTERNATIONAL CO., LTD., Taichung, Taiwan | 2025 - Present
• Lead and manage project planning, technical design and communication with industrial stakeholders
• Developed core software integrating LLM and MCP (Model Context Protocol) for intelligent navigation assistance
• Focus on AI-powered quality inspection and user guidance systems

Smart CNC Flexible Manufacturing Factory (Phase 1, 2 & 3)
IC STAR INDUSTRIAL CO., LTD., Taichung, Taiwan | 2024 - 2025
• Executed three-phase Smart CNC Flexible Manufacturing Factory project
• Redesigned Plant A and built Plant B, deploying 20 CNC machines and 10 robotic arms with full system integration
• Led a 42-member cross-disciplinary team covering robotics, CNC communication, software-hardware setup, and control design
• Achieved 80% reduction in manpower and significantly enhanced automation efficiency

Automated Robot Arm Forging Spraying System with AI-Based Path Generation (Phase 1 & 2)
IC STAR INDUSTRIAL CO., LTD., Changhua, Taiwan | 2023 - 2024
• Led project phases focusing on AI ant colony optimization planning and robotic control
• Achieved 27% higher production throughput, 91% defect reduction, and 21% cost savings across three forging lines
• Enabled flexible line switching with 83% shorter setup time and 97.3% trajectory matching accuracy

Automatic CNC Chip Entanglement Detection System
Industrial Technology Research Institute (ITRI), Taichung, Taiwan | 2023
• Developed an AI-based YOLO vision system to detect CNC tool chip entanglement in real time
• Implemented in unmanned smart manufacturing lines, reducing rework and maintenance costs

Development and Commercialization of Automatic Mushroom Stem-Cutting System
University Social Responsibility (USR) Project with Local Farmers, Xinshe, Taiwan | 2022 - Present
• Successfully mass-produced the smart mushroom stem-cutting machine, adopted by three manufacturers
• Two patents granted for the pressing and auto-sorting mechanisms
• Greatly improved cutting efficiency and quality in local agricultural industry

3D Vision-Guided Picking and Robotic System Integration Development
LIEBHERR-Verzahntechnik GmbH & PONGI Gear Expert CO., LTD., Taichung, Taiwan | 2022 - Present
• Integrated and optimized the Taiwan-distributed Liebherr LHRobotics.Vision random picking system
• Developed 3D vision-guided robotic coordination to enhance precision and flexibility
• Supported client implementation and application testing, provided technical solutions

HONORS & AWARDS

Awards During Bachelor's Program (2022-2025)

• 2025: Second Place - National Vocational and Technical College Student Practical Project Competition

• 2024: First Place - Intelligent Automation Equipment Invention Awards, 13th Edition

• 2024: Third Place - Taiwan Society of Precision Engineering's 113th Precision Engineering Project and Paper Award

• 2024: Honorable Mention - Mitsubishi Electric Contest of Automation, 1st Edition

• 2024: Honorable Mention - Mitsubishi Electric CNC Smart App Creative Development Competition

• 2023: Science and Technology Humanities Award - The 27th TDK Cup National Collegiate Creative Design and Production Competition

Early Achievements (2020-2022)

• Achieved 3 Gold, 2 Silver, and 4 Honorable Mentions in National Project and Science Fair Competitions

CONFERENCE PRESENTATIONS & EXHIBITIONS

2025 Taipei USR EXPO (September 2025)
Presented a smart mushroom stem-cutting system developed under the USR project, demonstrating sustainable agricultural automation and AI-based cutting control.

2024 Taipei International Automation Exhibition (August 2024)
Showcased an AI Automated Robot Arm Forging Spraying System solution and initiated collaboration with a local forging factory for potential joint development.

PROFESSIONAL CERTIFICATIONS & LICENSES

• Photovoltaic System Installation - Licensed Technician Class B (Ministry of Labor - WDA)

• AI Application Planner - Licensed Technician (Ministry of Economic Affairs - iPAS)

• Industrial Wiring - Licensed Technician Class C (Ministry of Labor - WDA)

• Commercial Wiring - Licensed Technician Class C (Ministry of Labor - WDA)

• Master of Computer Certificate - Class C (Chinese Computer Education Association)

INTERNATIONAL ACADEMIC EXCHANGE

SAKURA SCIENCE Exchange Program Member (2024)
Japan Science and Technology Agency (JST)
Suwa University of Science, Nagano, Japan
Host: Prof. Masahide Oshima (Vice President / Professor)

TECHNICAL SKILLS

Programming: Python, C#, C++, Matlab

AI & Vision: YOLO, Computer Vision, Machine Learning, AI Algorithm Development

Robotics: Robot Arm Programming & Control, Path Planning, System Integration

Automation: PLC Programming, CNC Communication, Industrial Control Systems

CAD/CAM: SolidWorks, 3D Modeling, Mechanical Design

LLM & AI: Large Language Models (LLM), Model Context Protocol (MCP), Generative AI

LANGUAGES

• Chinese - Native

• English - Proficient (TOEIC Score: 660, actively learning)

• German (Deutsch) - Beginner (actively learning)

SUMMARY OF KEY ACHIEVEMENTS

Research & Innovation:
• 2 journal papers (1 SCI-indexed), 2 patents, and 3 conference papers in intelligent automation and AI robotics
• Active contributor to academia-industry collaborative research

Industrial Impact:
• Led 6 major industry-academia collaboration projects
• Implemented large-scale automation: 20 CNC machines with 10 robot arms across two plants (three phases) and three forging lines (two phases)

• Achieved significant improvements: 80% manpower reduction, 27% production increase, 91% defect reduction

Recognition & Collaboration:
• Multiple national awards in automation innovation and system design
• Featured exhibitor at major national technology exhibitions
• International academic exchange participant (Japan)

REFERENCES

Available upon request

Last updated: February 2026